Lesson 1: Rational Expressions and Non-Permissible Values

Assignment

1. Is each a rational expression? Explain.

(a) $\frac{6}{a^2}$ (b) $\frac{\sqrt{t}+2}{t-5}$ (c) $\frac{5}{3^x}$ (d) x^2-4x+1

2. Evaluate $\frac{3x-1}{x^2-2x}$ for each value of x.

(a) $x=3$ (b) $x=-2$ (c) $x=\frac{1}{3}$ (d) $x=2$ (e) $x=0$

3. Evaluate $\frac{2a^2-b}{a+3b}$ for (a) $a=2$, $b=-1$ (b) $a=-3$, $b=1$.

4. (a) Complete the table for $\frac{2x}{x+1}$.

x	-1.1	-1.01	-1.001	-1	-0.999	-0.99	-0.9
$\frac{2x}{x+1}$							

(b) Describe the values near $x=-1$. Use multiplication to explain why the expression has no value at $x=-1$.

5. Determine the non-permissible values.

(a) $\frac{11}{6m}$ (b) $\frac{x-7}{x+9}$ (c) $\frac{2}{5-4y}$ (d) $\frac{x^2+3}{14}$

(e) $\frac{3p}{p^2+25}$ (f) $\frac{t+1}{(t-2)(3t+5)}$ (g) $\frac{8}{x^2y}$ (h) $\frac{w}{w(w-1)}$

6. Factor the denominator and determine the non-permissible values.

(a) $\dfrac{x+1}{x^2-x-12}$ (b) $\dfrac{5}{x^2-81}$ (c) $\dfrac{2a}{a^2+7a}$

(d) $\dfrac{1}{3x^2-10x-8}$ (e) $\dfrac{y-2}{4y^2+12y+9}$ (f) $\dfrac{x}{25-16x^2}$

(g) $\dfrac{6}{2x^2-8x-24}$ (h) $\dfrac{k+4}{k^3-4k^2+4k}$ (i) $\dfrac{x^2-9}{x^4-5x^2+4}$

7. Determine the non-permissible values for x in terms of y.

(a) $\dfrac{3x}{x-4y}$ (b) $\dfrac{x+y}{x^2-49y^2}$ (c) $\dfrac{1}{x^2+xy-6y^2}$

8. Determine the non-permissible values of $\dfrac{2}{x+6}-\dfrac{x}{x^2-4x}$.

9. (a) Complete the table.

x	-4	0	2	4	6
$\dfrac{x^2-4x}{x^2-16}$					
$\dfrac{x}{x+4}$					

(b) For which value of x in the table do the expressions differ? (c) State the NPVs of each expression.

10. Write a rational expression with a numerator of 1 and the denominator expanded that has the given NPVs.

(a) $x \neq 3, -8$ (b) $x \neq 0, -5$ (c) $x \neq \pm\frac{2}{3}$ (d) $x \neq \frac{1}{4}, -2$ (e) no NPVs

11. A group rents a cabin for \$960 and pays \$30 per person for food. The cost per person when n people go is $C = \dfrac{960 + 30n}{n}$ dollars.

(a) Determine the cost per person for 8 people.

(b) State the NPV and the values of n that make sense in this context.

(c) Use a calculator table (or try values) to find how many people must go for the cost to be \$90 per person.

12. A rectangle has an area of 60 cm^2 and a width of $(x - 3)$ cm, so its length is $\dfrac{60}{x - 3}$ cm.

(a) Determine the length when $x = 7$ and when $x = 15$.

(b) State the NPV. Which values of x make sense in this context? Why?

13. Jaden says the NPVs of $\dfrac{x - 5}{x^2 + 3x}$ are $x \neq 5$ and $x \neq -3$. Explain his error and give the correct NPVs.

14. The non-permissible values of $\dfrac{x + 2}{x^2 - 3x - 10}$ are

A. $x \neq 5$ only **B.** $x \neq -5, 2$ **C.** $x \neq -2, 5$ **D.** $x \neq -2$ only

15. Which expression has no non-permissible values?

A. $\dfrac{x}{x^2 - 1}$ **B.** $\dfrac{x^2 - 1}{x}$ **C.** $\dfrac{x - 1}{x^2 + 1}$ **D.** $\dfrac{1}{x^2 - 2x + 1}$

16. Which expression has the non-permissible values $x \neq 0, -\frac{3}{2}$?

A. $\dfrac{1}{2x^2 + 3x}$ **B.** $\dfrac{1}{2x^2 - 3x}$ **C.** $\dfrac{1}{3x^2 + 2x}$ **D.** $\dfrac{x}{2x + 3}$

17. Numerical Response The sum of the non-permissible values of $\dfrac{x + 7}{2x^2 - 13x + 15}$, to the nearest tenth, is _____.

18. Numerical Response The expression $\dfrac{4}{x^2 + kx + 18}$ has a non-permissible value of $x = 3$. Its other non-permissible value is _____.

Lesson 2: Simplifying Rational Expressions by Factoring

Assignment

1. Simplify. State the non-permissible values.

(a) $\dfrac{21x^4}{14x}$ (b) $\dfrac{-8m^3n^2}{12mn^5}$ (c) $\dfrac{4(a+1)}{10(a+1)^2}$ (d) $\dfrac{(y-6)(y+2)}{3(y+2)(y-1)}$

2. Explain each error.

(a) $\dfrac{x+10}{x+5} = 2$. Show with a value of x that the result is wrong.

(b) $\dfrac{5x-15}{5} = x-15$

(c) $\dfrac{x^2+x-6}{x^2-4} = \dfrac{x+3}{x+2}$, $x \neq -2$

3. Simplify. State the non-permissible values.

(a) $\dfrac{6x+18}{9}$ (b) $\dfrac{5a-20}{a^2-4a}$ (c) $\dfrac{x^2-49}{3x+21}$ (d) $\dfrac{8t^2+12t}{4t}$

4. Simplify. State the non-permissible values.

(a) $\dfrac{x^2+8x+15}{x^2+2x-15}$ (b) $\dfrac{m^2-36}{m^2-3m-18}$ (c) $\dfrac{x^2-10x+25}{x^2-25}$

(d) $\dfrac{2y^2-5y-3}{y^2-9}$ (e) $\dfrac{3x^2+5x-2}{6x^2-5x+1}$ (f) $\dfrac{4a^2-1}{2a^2+9a-5}$

5. Simplify. State the non-permissible values.

(a) $\dfrac{2x^2 - 2x - 40}{4x^2 - 64}$

(b) $\dfrac{x^3 - 9x}{x^3 + 6x^2 + 9x}$

(c) $\dfrac{6y^3 + 3y^2 - 18y}{12y^2 + 24y}$

(d) $\dfrac{x^4 - 16}{x^2 + x - 6}$

6. Simplify. State the non-permissible values (in (f), for a in terms of b).

(a) $\dfrac{3 - y}{y - 3}$

(b) $\dfrac{4x - 12}{3 - x}$

(c) $\dfrac{25 - a^2}{a^2 - 3a - 10}$

(d) $\dfrac{6 - x - x^2}{x^2 + 6x + 9}$

(e) $\dfrac{2x^2 - 7x + 6}{12 - 5x - 2x^2}$

(f) $\dfrac{(a - b)^2}{b^2 - a^2}$

7. Simplify, and state the non-permissible values for x in terms of y.

(a) $\dfrac{3x + 3y}{x^2 - y^2}$

(b) $\dfrac{x^2 + 4xy + 4y^2}{x^2 - 4y^2}$

(c) $\dfrac{2y - 6x}{9x^2 - y^2}$

(d) $\dfrac{x^2 - 3xy - 10y^2}{5y^2 - xy}$ (also state the NPV of y)

8. Which expressions are equivalent to $\dfrac{a+4}{a-1}$ (for their permissible values)? Explain each.

(a) $\dfrac{a^2+4a}{a^2-a}$ (b) $\dfrac{-a-4}{1-a}$ (c) $\dfrac{a+4}{1-a}$ (d) $\dfrac{a^2+3a-4}{(a-1)^2}$

9. Determine the missing numerator.

(a) $\dfrac{x+2}{x-5} = \dfrac{?}{x^2-25}$ (b) $\dfrac{3}{2x} = \dfrac{?}{8x^3}$ (c) $\dfrac{x-1}{4-x} = \dfrac{?}{x-4}$ (d) $\dfrac{2x}{x+3} = \dfrac{?}{x^2+x-6}$

(e) The expression $\dfrac{2x^2+kx-12}{x^2-16}$ simplifies to $\dfrac{2x-3}{x-4}$. Determine k.

10. A rectangle has an area of $(3x^2+10x-8)$ cm^2 and a length of $(x+4)$ cm.

(a) Write and simplify an expression for its width.

(b) The length is 3 cm more than the width. Determine x and the dimensions.

(c) Determine the exact area when $x = 1+\sqrt{2}$.

11. A triangle has an area of $(x^2+7x+12)$ cm^2 and a base of $(x+3)$ cm. Write and simplify an expression for its height.

12. A cylinder has radius r cm and height $(r+5)$ cm.

(a) Write the surface area $SA = 2\pi r^2 + 2\pi rh$ and the volume $V = \pi r^2 h$ in factored form.

(b) Write $\dfrac{V}{SA}$ in simplest form, state the NPVs, and evaluate it for $r = 5$.

13. Write a rational expression with a trinomial numerator and a trinomial denominator that simplifies to $\frac{x-2}{x+5}$ and has the NPVs $x \neq -5, 1$.

14. In simplest form, $\frac{x^2-4x}{16-x^2}$ is

A. $\frac{x}{x+4}$ **B.** $-\frac{x}{x+4}$ **C.** $\frac{x}{x-4}$ **D.** $-\frac{1}{4}$

15. Which expression is equivalent to $\frac{a-2}{3-a}$?

A. $\frac{2-a}{a-3}$ **B.** $\frac{a-2}{a-3}$ **C.** $-\frac{a+2}{a-3}$ **D.** $\frac{2}{3}$

16. When $\frac{2x^2-3x-2}{x^2-4}$ is written in simplest form, the result is

A. $\frac{2x+1}{x+2}$, $x \neq -2$

B. $\frac{2x+1}{x+2}$, $x \neq \pm 2$

C. $\frac{2x-1}{x+2}$, $x \neq \pm 2$

D. $\frac{2x+1}{x-2}$, $x \neq \pm 2$

17. Numerical Response The expression $\frac{3x^2-11x-4}{x^2-16}$ simplifies to $\frac{ax+1}{x+b}$. The value of $a+b$ is _____.

18. Numerical Response Simplify $\frac{x^2-5x-14}{x^2-49}$ and evaluate it at $x=3$. To the nearest tenth, the value is _____.

Lesson 3: Multiplying Rational Expressions

Assignment

1. Multiply. Write each answer in lowest terms.

(a) $\frac{21}{32} \times \frac{24}{35}$

(b) $-\frac{10}{27} \times \frac{18}{25}$

2. Simplify. State the non-permissible values.

(a) $\frac{9a^3b}{4c^2} \cdot \frac{10c^5}{21ab^4}$

(b) $\frac{-12x^2y^3}{35z} \cdot \frac{14z^4}{18x^5y}$

(c) $\frac{8m^4}{15n} \cdot \frac{-5n^3}{6m^2} \cdot \frac{9}{4mn}$

3. Simplify. State the non-permissible values.

(a) $\frac{12(x-5)}{x(x+2)} \cdot \frac{x^2(x+2)}{8(x-5)(x+1)}$

(b) $\frac{4a-20}{a^2+3a} \cdot \frac{5a+15}{2a-10}$

(c) $\frac{x^2-36}{9x} \cdot \frac{6x^3}{2x+12}$

(d) $(x^2-16) \cdot \frac{5x}{x^2+4x}$

4. Simplify. State the non-permissible values.

(a) $\dfrac{x^2+x-20}{x^2-3x-10}\cdot\dfrac{x^2+5x+6}{x^2+8x+15}$

(b) $\dfrac{y^2-49}{y^2+4y-21}\cdot\dfrac{y^2-9}{y^2-5y-14}$

(c) $\dfrac{2x^2-5x-3}{x^2-9}\cdot\dfrac{x^2+6x+9}{4x^2-1}$

(d) $\dfrac{6t^2+7t-3}{3t^2+12t}\cdot\dfrac{t^2+2t-8}{4t^2-9}$

5. Simplify. State the non-permissible values.

(a) $\dfrac{3-x}{x^2+4x}\cdot\dfrac{x^2-16}{2x-6}$

(b) $\dfrac{12-3m}{m^2+2m}\cdot\dfrac{m^2-4}{m^2-6m+8}$

(c) $\dfrac{25-x^2}{x^2-2x-15}\cdot\dfrac{x^2+3x}{x^2+5x}$

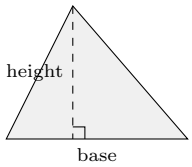
6. Simplify $\dfrac{a^2-a-6}{4a^2}\cdot\dfrac{2a^3+10a^2}{a^2-9}\cdot\dfrac{a^2+3a}{a^2+7a+10}$. State the non-permissible values.

7. Simplify. State the restrictions on the variables.

(a) $\dfrac{x^2-y^2}{3x+3y}\cdot\dfrac{6x}{x^2-2xy+y^2}$

(b) $\dfrac{a^2+ab-6b^2}{2a^2-8b^2}\cdot\dfrac{4a^2+8ab}{a^2+6ab+9b^2}$

8. A triangle has base $\dfrac{6x}{x^2-4}$ m and height $\dfrac{x^2+5x+6}{3x^2}$ m, where $x > 2$.



(a) Show that the area of the triangle simplifies to $\dfrac{x+3}{x(x-2)}$ square metres.

(b) Why must x be greater than 2 in this context?

(c) Determine the exact area when $x = 1+\sqrt{5}$.

9. A student simplified $\dfrac{x+6}{x^2-4}\cdot\dfrac{x-2}{x+3}$, $x \neq 2, -2, -3$, as shown. Identify the step with the error, explain it, and give the correct answer.

Step 1: $\dfrac{x+6}{(x-2)(x+2)}\cdot\dfrac{x-2}{x+3}$ Step 2: $\dfrac{x+6}{(x+2)(x+3)}$ Step 3: $\dfrac{\cancel{x+6}^{\,3}}{(\cancel{x+2})(x+3)} = \dfrac{3}{x+3}$

10. Determine the polynomial that belongs in the box: $\dfrac{x^2-x-6}{x^2+4x}\cdot\dfrac{\square}{x^2-9} = \dfrac{x+2}{x}$

11. For all permissible values of x, the product $\dfrac{x^2-25}{x^2+2x}\cdot\dfrac{x+2}{5-x}$ simplifies to

A. $-\dfrac{x+5}{x}$ **B.** $\dfrac{x+5}{x}$ **C.** $-\dfrac{x-5}{x}$ **D.** $\dfrac{x+5}{x(x+2)}$

12. The non-permissible values of $\dfrac{x+1}{x^2-9}\cdot\dfrac{x-3}{x^2+x}$ are

A. $x \neq 0, -3$
B. $x \neq 0, 3, -3$
C. $x \neq 0, -1, 3, -3$
D. $x \neq -1, 3, -3$

13. Numerical Response For all permissible values of x, the product $\dfrac{4x^2-36}{x^2+2x}\cdot\dfrac{3x^2+6x}{x^2+x-6}\cdot\dfrac{x-2}{x-3}$ simplifies to a whole number k. The value of k is _____.

14. Numerical Response Simplify $\dfrac{x^2-x-2}{x^2+3x}\cdot\dfrac{x^2-9}{x^2-4x+4}$, then evaluate it for $x=7$. To the nearest hundredth, the value is _____.

Lesson 4: Dividing Rational Expressions

Assignment

1. Simplify. Write each answer in lowest terms.

(a) $\frac{15}{16} \div \frac{25}{24}$

(b) $-\frac{14}{9} \div \frac{21}{6}$

(c) $\frac{5}{12} \times \frac{9}{20} \div \frac{3}{8}$

2. Simplify. State the non-permissible values.

(a) $\frac{8a^2b}{15c} \div \frac{4ab^3}{5c^2}$

(b) $\frac{-10x^3}{21y^2} \div \frac{15x}{14y^5}$

(c) $\frac{6m^2n}{5p} \cdot \frac{10p^3}{9mn^2} \div \frac{4p}{3n}$

3. State the non-permissible values. Do not simplify.

(a) $\frac{x-1}{x+6} \div \frac{x-8}{2x+3}$

(b) $\frac{x^2+5x}{x^2-49} \div \frac{x^2-3x-10}{x^2-x}$

4. Simplify. State the non-permissible values.

(a) $\frac{6x-18}{x^2+x} \div \frac{4x-12}{x+1}$

(b) $\frac{(2x+5)^2}{x^2-64} \div \frac{(2x+5)(x-3)}{x-8}$

(c) $\frac{9y^2-1}{y^2+3y} \div (3y+1)$

(d) $(x^2-2x) \div \frac{x^2-4}{5x}$

5. Simplify. State the non-permissible values.

(a) $\dfrac{x^2-3x-10}{x^2-4x+3} \div \dfrac{x^2+6x+8}{x^2+x-12}$

(b) $\dfrac{3a^2+5a-2}{a^2-4} \div \dfrac{9a^2-1}{a^2-2a}$

(c) $\dfrac{p^2-36}{p^2+p} \div \dfrac{p^2-5p-6}{p^2+3p+2}$

(d) $\dfrac{16-x^2}{2x^2+7x-4} \div \dfrac{x^2-2x-8}{4x^2-1}$

6. Simplify $\dfrac{x^2+xy-2y^2}{x^2-4y^2} \div \dfrac{x^2-y^2}{3x-6y}$. State the restrictions on the variables.

7. Simplify. State the non-permissible values.

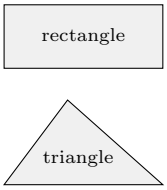
(a) $\dfrac{x^2-9}{x^2+2x} \div \dfrac{x-3}{x^2-4} \cdot \dfrac{x^2}{x^2+x-6}$

(b) $\dfrac{a^2-a-12}{a^2-16} \cdot \dfrac{a^2+4a}{a^2+6a+9} \div \dfrac{a^2-2a}{a+3}$

(c) $\dfrac{2y^2-8}{y^2+y} \div \dfrac{y^2+4y+4}{y+1} \div \dfrac{y-2}{3y}$

8. Simplify $\dfrac{\dfrac{4x^2-1}{x^2-3x}}{\dfrac{2x^2+x}{x^2-9}}$. State the non-permissible values.

9. A rectangle has length $\dfrac{x^2+4x}{x-3}$ cm and width $\dfrac{6}{x+4}$ cm. A triangle has base $\dfrac{x^2-9}{x}$ cm and height $\dfrac{4x}{x+3}$ cm. Here $x > 3$.



(a) Write and simplify an expression for each area.

(b) Write and simplify the ratio of the area of the rectangle to the area of the triangle.

(c) Evaluate the ratio when $x = 5$ and explain what it means.

10. Determine the rational expression that belongs in the box: $\dfrac{x^2-4}{x^2+3x} \div \square = \dfrac{x-2}{x}$

11. A student's work is shown. Identify both errors and give the correct answer.

$$\frac{x+1}{x-2} \div \frac{x+1}{x+5} = \frac{x-2}{x+1} \cdot \frac{x+1}{x+5} = \frac{x-2}{x+5}, \qquad x \neq 2, -5$$

12. The number of non-permissible values of $\dfrac{x^2-x-6}{x^2-4x} \div \dfrac{x^2+5x+6}{x^2-16}$ is

A. 3 **B.** 4 **C.** 5 **D.** 6

13. For $x \neq 0$, the expression $\frac{x}{3} \div \frac{x^2}{6} \cdot \frac{x}{2}$ is equal to

A. $\frac{x^2}{4}$ **B.** $\frac{4}{x^2}$ **C.** 1 **D.** $\frac{x^4}{36}$

14. For all permissible values of x, $\frac{x^2-1}{x+3} \div \frac{1-x}{x^2+3x}$ simplifies to

A. $-x(x+1)$ **B.** $x(x+1)$ **C.** $-\frac{1}{x(x+1)}$ **D.** $-x(x-1)$

15. Numerical Response For all permissible values of x, the complex fraction $\dfrac{\frac{3x^2+23x+40}{x^2+x-20}}{\frac{x+1}{x^2-3x-4}}$ simplifies to a linear expression $Ax + B$. The value of $A + B$ is _____.

Lesson 5: Adding and Subtracting with Monomial Denominators

Assignment

1. Simplify.

(a) $\frac{3}{4}+\frac{1}{6}$ (b) $\frac{5}{8}-\frac{1}{12}$ (c) $\frac{2}{5}+\frac{3}{10}-\frac{1}{4}$ (d) $\frac{1}{6}-\frac{7}{9}$

2. Simplify.

(a) $\frac{x+1}{2}+\frac{x-3}{5}$ (b) $\frac{2a+3}{4}-\frac{a-1}{3}$

(c) $\frac{3y-2}{6}-\frac{y+4}{9}$ (d) $\frac{m}{2}-\frac{m-5}{3}+\frac{m+1}{6}$

3. Simplify. State the non-permissible values.

(a) $\frac{4}{5x}+\frac{11}{5x}$ (b) $\frac{3a+7}{2a}-\frac{a-1}{2a}$ (c) $\frac{6t+1}{t-2}-\frac{4t+5}{t-2}$

(d) $\frac{x^2+2x}{x+5}-\frac{15}{x+5}$ (e) $\frac{y^2+4}{y-2}-\frac{4y}{y-2}$ (f) $\frac{5k}{3(k+1)}+\frac{5}{3(k+1)}$

4. Determine the LCD.

(a) $4x$ and $6x$ (b) $3a^2$ and $5a$ (c) $10mn^2$ and $4m^3$ (d) $2x$, $3y$ and $6xy$

5. Simplify. State the non-permissible values.

(a) $\frac{5}{2x} + \frac{1}{3x}$ (b) $\frac{7}{a} - \frac{2}{5a}$ (c) $\frac{3}{4y^2} + \frac{5}{6y}$ (d) $\frac{2}{9t} - \frac{5}{6t^3}$

(e) $\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}$ (f) $\frac{4}{3ab} - \frac{1}{2a^2}$ (g) $\frac{5}{6x^2y} + \frac{3}{8xy^3}$ (h) $\frac{7}{12c} - \frac{1}{4c} - \frac{1}{3c}$

6. Simplify. State the non-permissible values.

(a) $\frac{x+4}{2x} + \frac{x-1}{5x}$ (b) $\frac{3a-2}{4a} - \frac{a+6}{6a}$ (c) $\frac{2m+1}{3m} - \frac{m-4}{m^2}$

(d) $\frac{y-5}{6y^2} + \frac{y+2}{9y}$ (e) $\frac{2x-y}{xy} + \frac{x+3y}{x^2}$ (f) $\frac{5-3k}{10k^2} - \frac{2-k}{4k}$

7. Simplify. State the non-permissible values.

(a) $4 + \frac{3}{x}$ (b) $\frac{2}{a} - 5$ (c) $x + 1 - \frac{6}{x}$ (d) $\frac{2y}{3} - 1 + \frac{5}{4y}$

8. A student simplified $\frac{3x+1}{4x} - \frac{x-5}{4x}$ as $\frac{2x-4}{4x} = \frac{x-2}{2x}$. Find the error, give the correct answer, and confirm it by substituting $x = 1$.

9. Each statement is false. Show it is false by substituting the given values, and write a correct simplification.

(a) $\dfrac{1}{a}+\dfrac{1}{b}=\dfrac{2}{a+b}$ (use $a=2$, $b=2$)

(b) $\dfrac{x+6}{3x}=\dfrac{7}{3}$ (use $x=2$)

10. A rectangle has length $\dfrac{5}{2x}$ m and width $\dfrac{2}{3x}$ m, where $x>0$.

(a) Show that the perimeter is $\dfrac{19}{3x}$ m.

(b) Determine the perimeter when $x=0.25$, to the nearest hundredth of a metre.

11. A delivery driver travels 25 km on a highway at an average speed of $3s$ km/h, then 8 km in town at s km/h.

(a) Write the total driving time as a single rational expression in simplest form.

(b) Determine the total time, to the nearest minute, when $s=35$.

12. Determine the value of k so that $\dfrac{x+k}{2x}-\dfrac{x-3}{3x}$ simplifies to $\dfrac{1}{6}$ (for $x\neq 0$).

13. Write $\dfrac{7}{12x}$ in the form $\dfrac{a}{3x}+\dfrac{b}{4x}$, where a and b are positive integers.

14. For $x\neq 0$, $\dfrac{3}{4x}+\dfrac{1}{6x}$ is equal to

A. $\dfrac{4}{10x}$ **B.** $\dfrac{11}{12x}$ **C.** $\dfrac{11}{12x^2}$ **D.** $\dfrac{4}{12x}$

15. For $x\neq 0$, $\dfrac{2x-3}{5x}-\dfrac{x-4}{5x}$ simplifies to

A. $\dfrac{x-7}{5x}$ **B.** $\dfrac{x+1}{5x}$ **C.** $\dfrac{3x-7}{5x}$ **D.** $\dfrac{x+1}{10x}$

16. For $a, b \neq 0$, $\dfrac{a}{2b} + \dfrac{b}{3a}$ simplifies to

A. $\dfrac{a+b}{6ab}$ **B.** $\dfrac{a+b}{2b+3a}$ **C.** $\dfrac{3a^2+2b^2}{6ab}$ **D.** $\dfrac{2a^2+3b^2}{6ab}$

17. Numerical Response The expression $\dfrac{x+5}{3x} + \dfrac{x-2}{4x}$ can be written as $\dfrac{Ax+B}{12x}$. The value of $A+B$ is _____.

18. Numerical Response When $\dfrac{5}{2x} - \dfrac{1}{3x} + \dfrac{1}{4x}$ is written as a single expression $\dfrac{k}{12x}$, the value of k is _____.

Lesson 6: Adding and Subtracting with Binomial and Trinomial Denominators

Assignment

1. Simplify. State the non-permissible values.

(a) $\dfrac{2}{x+1}+\dfrac{3}{x-2}$ (b) $\dfrac{5}{a-4}-\dfrac{2}{a+3}$ (c) $\dfrac{4}{y}-\dfrac{1}{y-6}$

(d) $\dfrac{1}{2m-1}+\dfrac{1}{m+2}$ (e) $\dfrac{3}{3t+2}-\dfrac{2}{2t-1}$

2. Simplify. State the non-permissible values.

(a) $\dfrac{x+3}{x-1}+\dfrac{x-2}{x+1}$ (b) $\dfrac{a+2}{a-3}-\dfrac{a-1}{a+2}$ (c) $\dfrac{2x}{x+4}-\dfrac{x-3}{x-2}$

(d) $2+\dfrac{5}{x-3}$ (e) $\dfrac{x}{x+1}-1$

3. Factor the denominators first. Simplify fully and state the non-permissible values.

(a) $\dfrac{3}{4x+8}+\dfrac{1}{x+2}$

(b) $\dfrac{5}{x^2+2x}-\dfrac{2}{x}$

(c) $\dfrac{1}{6a-9}+\dfrac{2}{4a-6}$

(d) $\dfrac{y}{y^2-25}-\dfrac{1}{y+5}$

(e) $\dfrac{a}{2a+6}-\dfrac{9}{2a^2+6a}$

(f) $\dfrac{3}{x-2}-\dfrac{12}{x^2-4}$

4. Simplify fully. State the non-permissible values.

(a) $\dfrac{1}{x^2+3x+2}+\dfrac{1}{x^2+4x+3}$

(b) $\dfrac{4}{x^2-x-6}-\dfrac{3}{x^2-9}$

(c) $\dfrac{2}{a^2-a-2}+\dfrac{1}{a^2+a}$

(d) $\dfrac{5}{x^2+x-6}-\dfrac{1}{x^2+5x+6}$

(e) $\dfrac{3}{2x^2-x-1}-\dfrac{1}{x^2-x}$

5. Simplify. State the non-permissible values.

(a) $\dfrac{7}{a-2}+\dfrac{3}{2-a}$ (b) $\dfrac{x}{x-6}-\dfrac{6}{6-x}$ (c) $\dfrac{5}{x^2-1}+\dfrac{2}{1-x}$

6. Simplify each expression on its own first, then combine. State the non-permissible values.

(a) $\dfrac{x^2-4}{x^2+x-2}+\dfrac{3}{x-1}$ (b) $\dfrac{2t-6}{t^2-t-6}-\dfrac{1}{t+3}$

7. Show that $\dfrac{3}{x^2-4x-5}+\dfrac{1}{25-x^2}$ simplifies to $\dfrac{2(x+7)}{(x-5)(x+1)(x+5)}$, and state the non-permissible values.

8. Maya claims that $\dfrac{1}{x-1}-\dfrac{1}{x+1}=\dfrac{2}{x^2-1}$.

(a) Test her claim with $x=3$.

(b) One value does not prove the claim. Prove it algebraically.

9. A student's work is shown. Find the error and simplify correctly.

$$\frac{x+2}{x-3}-\frac{x-1}{x+3}=\frac{(x+2)(x+3)-(x-1)(x-3)}{(x-3)(x+3)}=\frac{x^2+5x+6-x^2-4x+3}{(x-3)(x+3)}=\frac{x+9}{(x-3)(x+3)}$$

10. A rectangle has length $\dfrac{4}{x-2}$ cm and width $\dfrac{2}{x+2}$ cm.

(a) Write its perimeter as a single expression in simplest form. Which values of x make sense?

(b) Determine the perimeter when $x = 4$.

11. A cyclist rides 24 km to a lake at an average speed of x km/h and rides back 4 km/h faster.

(a) Write the total riding time as a single rational expression in simplest form.

(b) Determine the total time when $x = 12$.

(c) Write a simplified expression for how much longer the trip to the lake takes than the trip back.

12. Extension. Simplify each complex fraction. State the non-permissible values.

(a) $\dfrac{\frac{1}{x} - \frac{1}{3}}{x - 3}$

(b) $\dfrac{2 - \frac{2}{x}}{1 - \frac{1}{x^2}}$

13. For $x \neq \pm 3$, $\dfrac{1}{x-3} - \dfrac{6}{x^2-9}$ simplifies to

A. $\dfrac{1}{x+3}$ **B.** $\dfrac{-5}{x^2-9}$ **C.** $\dfrac{x-9}{x^2-9}$ **D.** $\dfrac{1}{x-3}$

14. For $x \neq 5$, $\dfrac{2x}{x-5} + \dfrac{10}{5-x}$ is equal to

A. 2 **B.** $\dfrac{2x+10}{x-5}$ **C.** -2 **D.** $\dfrac{2x-10}{x^2-25}$

15. The non-permissible values of $\dfrac{3}{x^2-x-6} - \dfrac{1}{x^2+2x}$ are

A. $x \neq 3, -2$ **B.** $x \neq -3, 2, 0$ **C.** $x \neq -2, 0, 3$ **D.** $x \neq 0, 2$

16. Numerical Response For $x \neq \pm 1$, $\dfrac{4}{x-1} + \dfrac{1}{x+1} - \dfrac{8}{x^2-1}$ simplifies to $\dfrac{k}{x+1}$. The value of k is _____.

17. Numerical Response When simplified, $\dfrac{2}{x^2-x-6}+\dfrac{3}{x^2-4}$ can be written as $\dfrac{Ax+B}{(x+2)(x-3)(x-2)}$, where A and B are integers. The value of $A-B$ is _____.

Lesson 7: Solving Rational Equations

Assignment

1. State the non-permissible values.

(a) $\dfrac{5}{x-8} = \dfrac{1}{x}$

(b) $\dfrac{x}{2x+5} + 3 = \dfrac{1}{x-4}$

(c) $\dfrac{2}{x^2-49} = \dfrac{1}{x+7}$

(d) $\dfrac{4}{x^2-3x-10} + \dfrac{1}{x+2} = 0$

2. Verify whether the given value is a solution of the equation.

(a) $x = 6$; $\dfrac{4}{x-2} + \dfrac{1}{2} = \dfrac{9}{x}$

(b) $x = 3$; $\dfrac{x+3}{x-1} = \dfrac{2x}{x+3}$

3. State the NPVs, then solve.

(a) $\dfrac{6}{x} - 1 = \dfrac{2}{x}$

(b) $\dfrac{3}{2x} + \dfrac{1}{4} = \dfrac{5}{x}$

(c) $\dfrac{x-3}{3x} = \dfrac{1}{6} + \dfrac{2}{x}$

(d) $\dfrac{5}{a} - \dfrac{2a+1}{4a} = \dfrac{1}{2}$

4. State the NPVs, then solve.

(a) $\dfrac{3}{x+4} = \dfrac{5}{x}$

(b) $\dfrac{2x-1}{x+3} = \dfrac{3}{4}$

(c) $\dfrac{3m+2}{m-1} = \dfrac{6m-1}{2m+3}$

(d) Solve for y: $\dfrac{4}{5-2y} = -3$

5. State the NPVs, solve and verify.

(a) $\dfrac{1}{x+5} + \dfrac{3}{x-5} = \dfrac{10}{x^2-25}$

(b) $\dfrac{4}{x-2} - \dfrac{3}{x+1} = \dfrac{15}{x^2-x-2}$

(c) $\dfrac{2}{x^2+3x} = \dfrac{5}{x+3} - \dfrac{1}{x}$

6. Solve. (Watch for denominators that are opposites.)

(a) $\dfrac{x}{x-6} + \dfrac{2}{6-x} = 5$

(b) $\dfrac{3}{2x-1} = \dfrac{5}{1-2x} + 4$

7. Solve. The x^2 terms cancel.

(a) $\dfrac{x}{x+4} + \dfrac{2}{x} = 1$

(b) $\dfrac{2x}{x+1} - \dfrac{x}{x-2} = \dfrac{x^2+4}{x^2-x-2}$

8. Show that each equation has no solution.

(a) $\dfrac{x+1}{x} - \dfrac{x}{x-2} = \dfrac{1}{x}$

(b) $\dfrac{3x}{x+1} - \dfrac{2x}{x-1} = \dfrac{x^2+5}{x^2-1}$

9. Use a graphing calculator to solve each equation, to the nearest hundredth. State the window you used.

(a) $\dfrac{3}{x-2} = x+1$

(b) $\dfrac{x}{x+1} = \dfrac{4}{x} - 1$

10. Kai solved $\dfrac{5}{x} + 2 = \dfrac{9}{x}$ like this: "Multiply by x: $5 + 2 = 9$. Since $7 \neq 9$, there is no solution." Explain the error and solve the equation correctly.

11. The equation $\dfrac{k}{x-1} + \dfrac{2}{x} = 3$ has the solution $x = 2$. Determine the value of k.

12. At a school concert, 85 more student tickets than adult tickets were sold. The ratio of adult tickets to student tickets was $2 : 7$. Write and solve a rational equation to find how many of each type were sold.

13. If $\dfrac{3}{2-y} = 4$, then y equals

A. $-\frac{5}{4}$ **B.** $\frac{11}{4}$ **C.** $\frac{5}{4}$ **D.** 5

14. The non-permissible values of $\dfrac{2}{x^2-4x} = \dfrac{1}{x+1}$ are

A. $0, 4$ **B.** $4, -1$ **C.** $0, 4, -1$ **D.** $0, -4, 1$

15. Multiplying every term of $\dfrac{3}{x} - \dfrac{1}{2} = \dfrac{5}{4x}$ by the LCD gives

A. $3 - 1 = 5$ **B.** $12 - x = 5$ **C.** $12 - 2x = 5$ **D.** $12x - 2x = 5x$

16. Numerical Response The solution of $\dfrac{4}{x} + \dfrac{3x}{x+5} = 3$, to the nearest hundredth, is _____.

17. Numerical Response The solution of $\frac{2}{x-1} + \frac{3}{x+1} = \frac{12}{x^2-1}$, to the nearest tenth, is _____.

Lesson 8: Rational Equations That Lead to Quadratics

Assignment

1. State the NPVs and solve.

(a) $x - \dfrac{12}{x} = 4$ (b) $\dfrac{5}{x} + \dfrac{x}{2} = \dfrac{7}{2}$ (c) $\dfrac{2}{x^2} - \dfrac{1}{x} = 1$ (d) $1 + \dfrac{9}{x^2} = \dfrac{6}{x}$

2. State the NPVs, solve and verify.

(a) $\dfrac{x-2}{3} = \dfrac{8}{x}$ (b) $\dfrac{2x}{x+3} = \dfrac{x+2}{x}$

3. State the NPVs and solve. Reject any extraneous root.

(a) $\dfrac{x}{x+2} + \dfrac{4}{x-1} = \dfrac{12}{x^2+x-2}$ (b) $\dfrac{x}{x+4} + \dfrac{3}{x-4} = \dfrac{32}{x^2-16}$

4. Consider $\dfrac{x^2+3x-4}{x+4} = 3x - 5$.

(a) Solve by cross-multiplying.

(b) Show a second method: simplify the left side first, then solve. Which method avoids the extraneous root?

5. Show that $\dfrac{2x}{x-2} - \dfrac{8}{x^2-2x} = \dfrac{4}{x}$ has no solution.

6. Solve. Give exact roots in simplest radical form, then round to the nearest hundredth.

(a) $\dfrac{4}{x} - 3 = \dfrac{x}{2}$

(b) $\dfrac{2}{x+1} + \dfrac{1}{x-2} = 1$

(c) Use a graphing calculator to confirm your answers to (b).

7. Use the discriminant to show that $\dfrac{x}{x-1} = \dfrac{2}{x} - 1$ has no real solution.

8. Solve $\dfrac{x}{x-5} + \dfrac{3}{5-x} = \dfrac{x}{2}$.

9. Priya solved $\dfrac{6}{x} = x - 1$ by writing "$6 = x - 1$, so $x = 7$." Explain her error, then solve the equation correctly.

10. One root of $x + \dfrac{k}{x} = 7$ is $x = 2$. Determine k and the other root.

11. For a thin lens, $\dfrac{1}{f} = \dfrac{1}{p} + \dfrac{1}{q}$, where f is the focal length, p is the distance from the object to the lens and q is the distance from the lens to the image (all in cm). A lens has $f = 12$ cm, and the image is 32 cm farther from the lens

than the object is. Determine p and q.

12. The solution of $\dfrac{x^2-9}{x-3}=2x$ is

A. $x=3$ **B.** $x=3$ or $x=-3$ **C.** $x=-3$ **D.** no solution

13. The solution(s) of $\dfrac{a}{a-2}-\dfrac{4}{a+2}=\dfrac{8}{a^2-4}$ is (are)

A. $a=0$ and $a=2$ **B.** $a=0$ only **C.** $a=2$ only **D.** no solution

14. The roots of $\dfrac{x}{x+1}=\dfrac{2}{x}$ are

A. $-1\pm\sqrt{3}$ **B.** $2\pm2\sqrt{3}$ **C.** $1\pm\sqrt{3}$ **D.** $1\pm\sqrt{5}$

15. Numerical Response The positive root of $\dfrac{x+1}{x-2}=\dfrac{3x}{x+2}$, to the nearest hundredth, is _____.

16. Numerical Response The solution of $\dfrac{x}{x-3}-\dfrac{13}{x+3}=\dfrac{18}{x^2-9}$ is $x=$ _____.

Lesson 9: Problems Involving Rational Equations (work, motion, numbers)

Assignment

1. (a) The sum of a number and its reciprocal is $\frac{29}{10}$. Determine the number.

(b) One number is 6 more than another. The reciprocal of the smaller number minus the reciprocal of the larger number is $\frac{3}{20}$. Determine the numbers.

2. The numerator of a fraction is 4 less than its denominator. If the numerator is tripled and the denominator is increased by 6, the new fraction is equal to 1. Determine the original fraction.

3. A train travels 300 km in the same time that a car travels 210 km. The train's average speed is 30 km/h greater than the car's. Determine each speed and the time taken.

4. Priya cycles 60 km. If she rode 5 km/h faster, the trip would take 1 hour less.

(a) Complete the table, using s for her actual speed.

	Distance (km)	Speed (km/h)	Time (h)
Actual			
Faster			

(b) Write an equation and determine her actual speed and time.

5. A family drives 225 km to a lake. On the way home, in heavy traffic, they average 15 km/h less and the trip takes 30 minutes longer. Determine the average speed and the time for each direction.

6. Noah runs 12 km and then walks 3 km. His running speed is 4 km/h greater than his walking speed, and the whole route takes 2.5 h. Determine his walking and running speeds, to the nearest tenth of a km/h.

7. A canoeist who paddles at 6 km/h in still water goes 12 km downstream and returns to the start in a total of 4.5 h. Determine the speed of the current.

8. With a 30 km/h tailwind, a small plane flies 450 km in the same time it flies 330 km into the same wind as a headwind. Determine the plane's airspeed (its speed in still air) and the time for each flight.

9. (a) One printer can print the yearbooks in 30 min and another in 45 min. How long will the job take using both?

(b) Working together, two volunteers can sort a food-bank delivery in 4 h. One of them alone needs 6 h. How long would the other need alone?

10. A new machine can package an order 3 h faster than an old machine. Using both machines, the order takes 2 h. How long does each machine take alone?

11. (a) Ali can tile a floor in 9 h and Sam can tile it in 6 h. Ali works alone for 4 h and then Sam joins him. How much longer does the job take?

(b) A hot tub fills in 30 min with the drain closed and empties in 45 min with the tap off. How long does it take to fill if the drain is left open?

12. A hockey team has won 9 of its first 20 games. How many games in a row must it win to raise its winning percentage to at least 60%? Explain your answer.

13. If you drive a distance at speed a and return the same distance at speed b, your average speed for the round trip is $\frac{2ab}{a+b}$. Zoe drives to a city at 60 km/h in heavy traffic. How fast must she drive home to average 75 km/h for the round trip?

14. Liam drives 200 km in the same time that Noor drives 240 km. Noor's average speed is 12 km/h greater than Liam's. If Liam's average speed is s km/h, an equation that can be used to determine s is

A. $\frac{200}{s+12} = \frac{240}{s}$ **B.** $200s = 240(s+12)$ **C.** $\frac{200}{s} = \frac{240}{s+12}$ **D.** $\frac{200}{s} + \frac{240}{s+12} = 12$

15. Pat can do a job alone in x hours. The fraction of the job Pat completes in 3 hours is

A. $\frac{3}{x}$ **B.** $\frac{x}{3}$ **C.** $3x$ **D.** $\frac{1}{3x}$

16. One pipe fills a tank in 3 h and a second pipe fills it in 6 h. With both pipes open, the tank fills in

A. 4.5 h **B.** 9 h **C.** 2 h **D.** 1.5 h

17. Numerical Response One garden hose fills a pond in 60 min and a second hose fills it in 40 min. Using both hoses, the number of minutes to fill the pond is _____.

Lesson 10: Unit 6 Review

Assignment

1. State the non-permissible values.

(a) $\dfrac{x-4}{5x^2}$ (b) $\dfrac{2x+1}{x^2-49}$ (c) $\dfrac{x}{3x^2-5x-2}$ (d) $\dfrac{x+3}{x^2+4}$

(e) Evaluate $\dfrac{x^2-1}{x+3}$ for $x=2$, and explain why it cannot be evaluated for $x=-3$.

2. Simplify and state the non-permissible values.

(a) $\dfrac{18a^3b}{24ab^4}$ (b) $\dfrac{x^2+6x+9}{x^2-9}$

(c) $\dfrac{6-2x}{x^2-7x+12}$ (d) $\dfrac{4x^2-9}{6x^2+5x-6}$

3. For $x \neq 5$, the expression $\dfrac{x-5}{5-x}$ is equal to

A. -1 **B.** 1 **C.** 0 **D.** $x-5$

4. (a) Kim wrote $\dfrac{x+6}{x+3}=2$. Explain the error. Test your explanation with $x=1$.

(b) Raj simplified $\dfrac{x^2-4}{x^2-2x}$ to $\dfrac{x+2}{x}$ and wrote "$x \neq 0$". What is missing, and why?

5. Simplify and state the non-permissible values.

(a) $\dfrac{15x^2}{8y} \cdot \dfrac{4y^3}{5x}$ (b) $\dfrac{x^2-x-20}{x^2-16} \cdot \dfrac{x-4}{2x-10}$

6. Which expression is equivalent to $\dfrac{x+2}{x-3}$ for $x \neq -1, 3$?

A. $\dfrac{x+4}{x-1}$ **B.** $\dfrac{2x+2}{2x-3}$ **C.** $\dfrac{x^2+3x+2}{x^2-2x-3}$ **D.** $\dfrac{x^2+2}{x^2-3}$

7. Simplify and state the non-permissible values.

(a) $\dfrac{x^2-1}{x^2+3x} \div \dfrac{x+1}{x^2-9}$

(b) $\dfrac{3x^2+5x-2}{x^2+4x+4} \div \dfrac{9x^2-1}{x^2+2x}$

8. The non-permissible values of $\dfrac{x+2}{x-5} \div \dfrac{x-3}{x+1}$ are

A. $5, -1$ only **B.** $5, 3$ only **C.** $-2, 5, 3, -1$ **D.** $5, -1, 3$

9. Simplify and state the non-permissible values.

(a) $\dfrac{3}{4x} + \dfrac{5}{6x}$ (b) $\dfrac{2}{3a^2b} - \dfrac{5}{6ab^2}$ (c) $\dfrac{x+2}{2x} - \dfrac{x-3}{5x}$

10. Numerical Response The sum $\dfrac{3}{x} + \dfrac{2}{x+1}$ can be written as $\dfrac{ax+b}{x(x+1)}$. Write the value of a in the first box and the value of b in the second box.

11. Simplify and state the non-permissible values.

(a) $\dfrac{4}{x-2}+\dfrac{3}{x+5}$

(b) $\dfrac{x}{x^2-5x+6}-\dfrac{2}{x^2-4}$

(c) $\dfrac{5}{x-3}+\dfrac{2}{3-x}$

12. In simplest form, $\dfrac{2}{x}-\dfrac{3}{x^2}$ is

A. $-\dfrac{1}{x^2-x}$ **B.** $-\dfrac{1}{x}$ **C.** $\dfrac{2x-3}{x^2}$ **D.** $\dfrac{2-3x}{x^2}$

13. Simplify $\dfrac{x^2-4}{x^2+x}\cdot\dfrac{x+1}{x-2}\div\dfrac{x+2}{3x}$ and state the non-permissible values.

14. Numerical Response $\dfrac{3}{x-4}=\dfrac{ax+b}{x^2-2x-8}$ for all permissible values of x. Write the value of a in the first box and the value of b in the second box.

15. State the non-permissible values and solve.

(a) $\dfrac{4}{x}-\dfrac{1}{3}=\dfrac{2}{x}$

(b) $\dfrac{x+1}{x-2}=\dfrac{x-3}{x+4}$

(c) $\dfrac{3}{x-1}+\dfrac{2}{x+1}=\dfrac{10}{x^2-1}$

16. When the equation $\dfrac{3}{x-2}+\dfrac{x}{x+2}=2$ is solved by multiplying every term by the LCD, an equation that appears is

A. $x^2+x+4=0$ **B.** $x^2-x-14=0$ **C.** $x^2-x+2=0$ **D.** $x^2+x-14=0$

17. State the non-permissible values, solve and verify.

(a) $x - \dfrac{12}{x} = 1$

(b) $\dfrac{2x}{x+3} + \dfrac{1}{x-1} = \dfrac{4}{x^2+2x-3}$

18. The extraneous root of $\dfrac{x^2}{x-4} = \dfrac{16}{x-4}$ is

A. 4 **B.** -4 **C.** 16 **D.** 0

19. Numerical Response The positive root of $\dfrac{5}{x} + \dfrac{3}{x+2} = 2$, to the nearest hundredth, is _____.

20. The difference between a number and 6 times its reciprocal is 1. Determine the number.

21. A boat that travels at 20 km/h in still water goes 36 km downstream in the same time it goes 24 km upstream. Determine the speed of the current and the time for each trip.

22. Sara drives 150 km to a cabin at s km/h. If she drove 25 km/h faster, the trip would take 30 minutes less. An equation that can be used to determine s is

A. $\dfrac{150}{s} - \dfrac{150}{s+25} = 30$

B. $\dfrac{150}{s} - \dfrac{150}{s+25} = \dfrac{1}{2}$

C. $\dfrac{150}{s+25} - \dfrac{150}{s} = \dfrac{1}{2}$

D. $\dfrac{150}{s} + \dfrac{150}{s+25} = \dfrac{1}{2}$

23. Solve the equation from the previous question to find Sara's speed and her driving time.

24. Kai can wash the team's jerseys in 3 h. With Mo helping, the job takes 1 h 12 min. How long would Mo take alone?

25. Numerical Response One pump takes 9 h less than a second pump to drain a pond. Together they drain it in 6 h. The number of hours the faster pump takes alone is _____.

26. Lee can do a job in 4 h and Pat can do it in 6 h. Working together for 1 h, they complete

A. $\frac{1}{10}$ of the job **B.** $\frac{1}{5}$ of the job **C.** $\frac{2}{5}$ of the job **D.** $\frac{5}{12}$ of the job

27. A baseball player has 18 hits in 60 at-bats. How many hits in a row would raise her batting average ($\frac{\text{hits}}{\text{at-bats}}$) to 0.400?